Part Number: STM32F405RG

Manufacturer: STMicroelectronics | Packages: LQFP-64, VQFN-48, WLCSP-90

DESCRIPTION

The STM32F405RG is a high-performance ARM Cortex-M4 32-bit RISC microcontroller operating at up to 168 MHz with FPU, 1 MB Flash memory, 192 KB SRAM, and rich industrial communication peripherals.

KEY FEATURES

- Core: ARM 32-bit Cortex-M4 CPU with FPU, frequency up to 168 MHz
- Memories: 1024 KB Flash memory and 192 KB SRAM (including 64 KB CCM data RAM)
- Clock, reset and supply management: 1.8 V to 3.6 V application supply and I/Os
- 3x 12-bit ADC (2.4 MSPS) up to 24 channels and 2x 12-bit D/A converters
- 3x I2C (400 kHz Fast-Mode), 3x SPI (42 MHz), 4x UART (10.5 Mbps), 2x CAN (1 Mbps), 1x USB 2.0 OTG FS
- Operating Temperature Range: -40 °C to +85 °C | Storage Temperature Range: -65 °C to +150 °C
- Maximum Junction Temperature: +125 °C | Theta_JA: 45.0 °C/W | Theta_JC: 12.0 °C/W

ELECTRICAL CHARACTERISTICS

Symbol	Parameter	Min	Typ	Max	Unit	Conditions	Category
VDD_MAX	Maximum Supply Voltage	-0.3	-	4.0	V	Absolute maximum rating	Absolute Maximum
VIN_MAX	Maximum Input Pin Voltage	-0.3	-	4.0	V	Non-5V tolerant pins	Absolute Maximum
VDD	Operating Supply Voltage	1.8	3.3	3.6	V	Full speed operation	Operating
VDDA	Analog Supply Voltage	1.8	3.3	3.6	V	ADC active	Operating
VIL	Input Low Level Voltage	-0.3	0.0	0.8	V	VDD = 3.3V	Operating
VIH	Input High Level Voltage	2.0	3.3	3.6	V	Non-FT GPIO, VDD=3.3V	Operating
IDD	Supply Current Run Mode	15.0	87.0	138.0	mA	fHCLK = 168 MHz, peripherals on	Operating
FCLK	System Clock Frequency	0.0	168.0	168.0	MHz	PLL driven from HSE	Operating
VREF	ADC Reference Voltage	1.8	3.3	3.6	V	Low-noise reference required	Operating

PIN CONFIGURATION

Pin	Name	Type	Voltage Domain	Description
1	VBAT	Power	1.65V-3.6V	Backup battery supply for RTC and backup registers
7	NRST	Input	3.3V	External active-low system reset input with internal pull-up
14	PA0	I/O	3.3V only	General purpose I/O, ADC1_IN0, TIM2_CH1, WKUP
29	PB6	I/O	5V tolerant (FT)	I2C1_SCL, TIM4_CH1, USART1_TX
30	PB7	I/O	5V tolerant (FT)	I2C1_SDA, TIM4_CH2, USART1_RX
41	PA5	I/O	3.3V only	SPI1_SCK, ADC1_IN5, DAC_OUT2
64	VDD	Power	3.3V	Digital power supply rail

SILICON ERRATA & TROUBLESHOOTING NOTES

	Symptom	Workaround
Bus Busy Flag Lockup	I2C1 BUSY flag remains stuck high after master reset or SDA glitch	Toggle SCL 9 clock pulses via GPIO open-drain mode and set SWR
1 Switching Noise on PA5	High ADC conversion jitter when SPI1_SCK toggles concurrently on PA5	Disable SPI1 clock during precision 12-bit ADC sampling or add RC